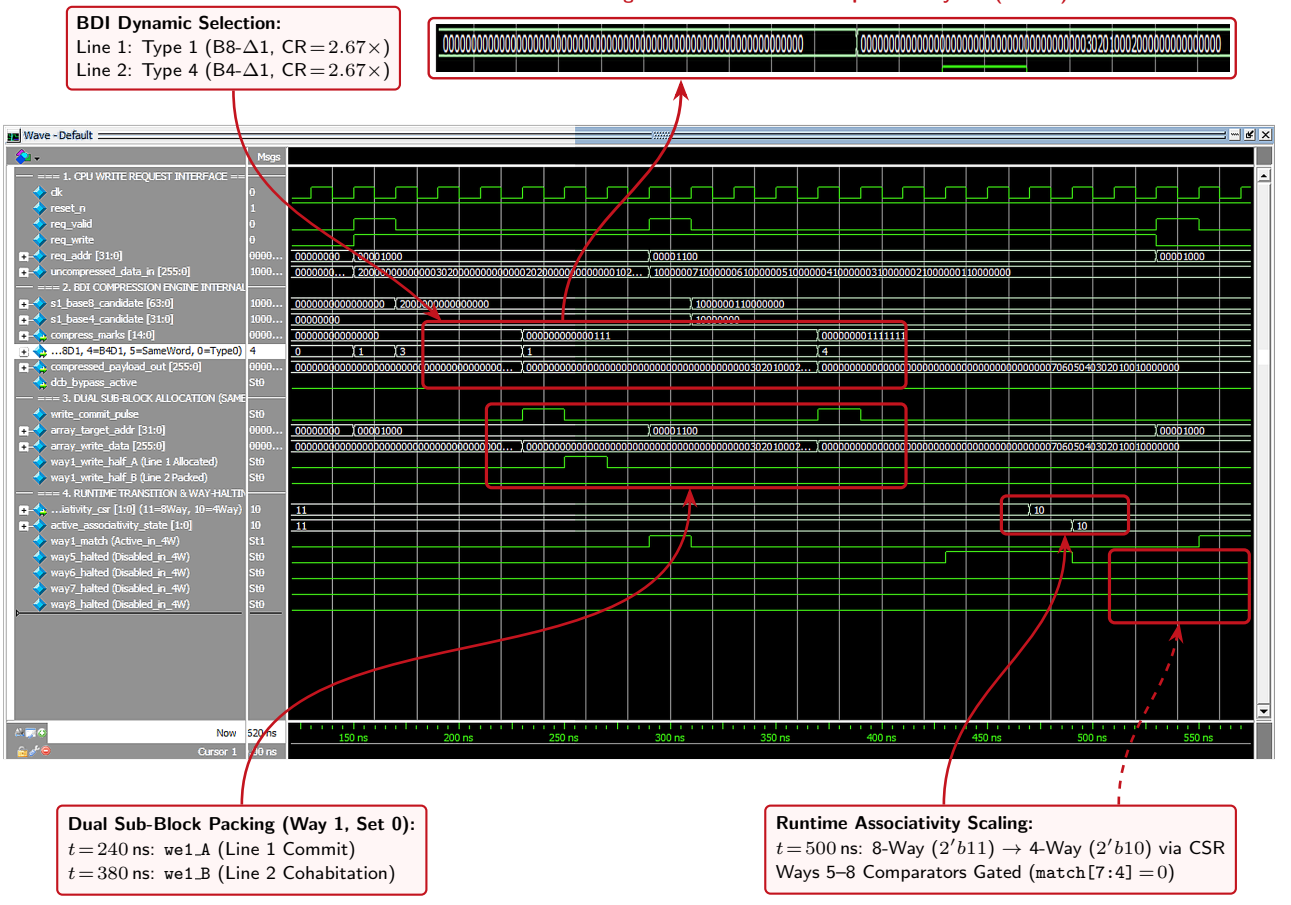

Magnified View: 256-Bit Compressed Payload (B8-Δ1)
BDI Dynamic Selection:
Line 1: Type 1 (B8-Δ1, CR = 2.67×)
Line 2: Type 4 (B4-Δ1, CR = 2.67×)
Wave - Default
Msgs
=== 1. CPU WRITE REQUEST INTERFACE ===
clk
reset_n
req_valid
req_write
req_addr [31:0]
uncompressed_data_in [255:0]
=== 2. BDI COMPRESSION ENGINE INTERNAL
s1_base8_candidate [63:0]
s1_base4_candidate [31:0]
compress_marks [14:0]
...8D1, 4=B4D1, 5=SameWord, 0=Type0)
compressed_payload_out [255:0]
dcb_bypass_active
=== 3. DUAL SUB-BLOCK ALLOCATION (SAME
write_commit_pulse
array_target_addr [31:0]
array_write_data [255:0]
way1_write_half_A (Line 1 Allocated)
way1_write_half_B (Line 2 Packed)
=== 4. RUNTIME TRANSITION & WAY-HALTIN
...iativity_csr [1:0] (11=8Way, 10=4Way)
active_associativity_state [1:0]
way1_match (Active_in_4W)
way5_halted (Disabled_in_4W)
way6_halted (Disabled_in_4W)
way7_halted (Disabled_in_4W)
way8_halted (Disabled_in_4W)
Now
Cursor 1
150 ns
200 ns
250 ns
300 ns
350 ns
400 ns
450 ns
500 ns
550 ns
Dual Sub-Block Packing (Way 1, Set 0):
t = 240 ns: we1_A (Line 1 Commit)
t = 380 ns: we1_B (Line 2 Cohabitation)
Runtime Associativity Scaling:
t = 500 ns: 8-Way (2′b11) → 4-Way (2′b10) via CSR
Ways 5–8 Comparators Gated (match[7:4] = 0)